

7.07 The Simplex Algorithm			
7.07a	Use a simplex tableau		a) Be able to set up an initial simplex tableau in standard format. <i>Rows to show objective to be maximised, followed by constraints. Columns to represent objective, variables and then slack variables, with the column representing the right-hand side as the last column.</i>
7.07b			b) Be able to perform an iteration of a simplex tableau, including the choice of the pivot. <i>The column with the most negative value in the objective row should be chosen, unless other specific instruction is given.</i>
7.07c			c) Be able to interpret the values of the variables, slack variables and objective at any stage and know when the optimum has been achieved. <i>Includes discussing the effect of changes to the coefficients.</i>
7.07d	Terminology		d) Be able to use the terms “basic feasible solution”, “basic variable” and “non-basic variable” appropriately. <i>Basic variables correspond to columns consisting of zeroes together with a single 1, the other variables are non-basic.</i>
7.07e	Graphical and algebraic interpretation of iterations		e) Understand what an iteration of the simplex algorithm achieves, in terms of moving along edges of a multi-dimensional convex polygon, usually in two or three dimensions, and be able to apply this to problems.
7.07f			f) Be able to explain, algebraically, some of the calculations used in the simplex algorithm.

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
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